

DOLLY POS — DATABASE ARCHITECTURE & SCHEMA MANUAL

Complete Technical Reference • Entity-Relationship Diagrams • Table Blueprints • Foreign Key Integrity

Target System: Dolly POS v1.0.0	Store: Dolly Toys & Kids Wear, Dhule	Date: September 2026
Primary Database: PostgreSQL 16 (64-bit)	Offline Fallback: SQLite 3 Zero-Config	Total Tables: 23 Relational Tables
ORM Framework: SQLAlchemy 2.0 (FastAPI)	Foreign Key Constraints: 19 Active Relations	Integrity: ACID Compliant • Cascade Safe

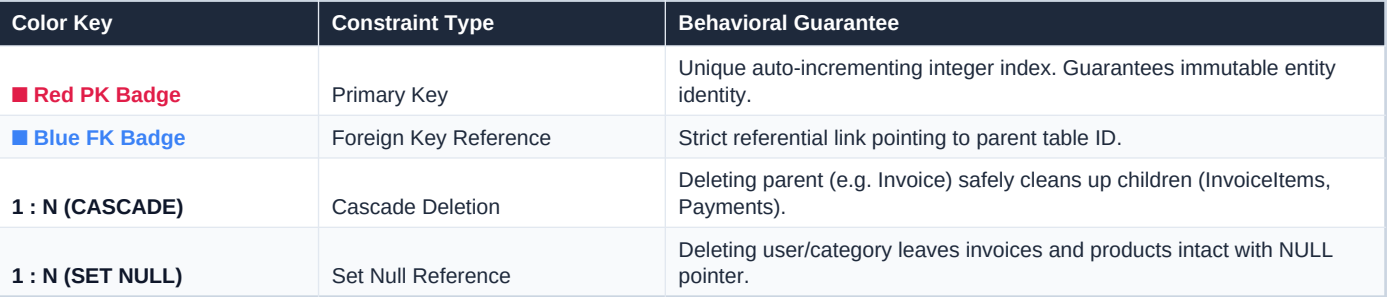
1. Executive Database Overview

The Dolly POS database architecture is engineered for **ultra-fast point-of-sale transactional throughput**, rock-solid inventory reconciliation, double-entry financial integrity for customer khata & vendor dues, and seamless real-time synchronization between PostgreSQL 16 (production) and local SQLite (offline fallback).

All tables strictly adhere to 3rd Normal Form (3NF) while embedding selective denormalizations (such as `purchase_rate_snapshot` on bill items and `customer_name` on invoices) to guarantee that historical financial accounting records, daily P&L; audits, and tax invoices remain immutable even if product prices or master records are modified years later.

Cluster	Primary Tables	Core Business Responsibility
1. Authentication & Security	users, audit_logs	RBAC permissions (Owner/Staff), Bcrypt password hashes, and immutable audit logs.
2. Master Catalog & Inventory	categories, subcategories, products, product_price_history	Barcode scan lookups, multi-attribute stock (size/color), MRP/Purchase rates, and price change audit trail.
3. Billing & Multi-Tender POS	invoices, invoice_items, payments	Lightning fast checkout, split payments (Cash+UPI+Card), 5-rule MDR charge engine, and thermal formatting.
4. Customer Khata Ledger	customers, customer_ledger	Double-entry credit ledger, running balances, credit limit verification, and WhatsApp bill messaging.
5. Vendor Procurement	vendors, vendor_ledger, purchases, purchase_items, procurement_notes	Stock replenishment, supplier invoice scanning, vendor payables ledger, and F10 Quick Buying Wishlist.
6. Returns & Restocking	return_orders, return_items	Original bill verification, restock vs. damaged inventory routing, and customer refund/exchange processing.
7. Expenses & Settings	expenses, store_settings, whatsapp_logs, lost_demand_logs	Shop operational overheads, thermal printer configs, social branding, and AI reorder tracking.

The following diagram illustrates the complete relational topology of the Dolly POS database. Boxes indicate relational entities with Primary Keys (**PK** in red) and Foreign Keys (**FK** in blue). Connecting lines denote active relationship cardinalities and referential integrity constraints.



3. Cluster Deep-Dive: Master Catalog & Product Inventory

This cluster governs product master definitions, hierarchical taxonomy (Categories & Subcategories), barcode indexing for sub-millisecond laser scanner checkout, live stock quantity counters, damaged goods diversion, and automated price audit tracking.

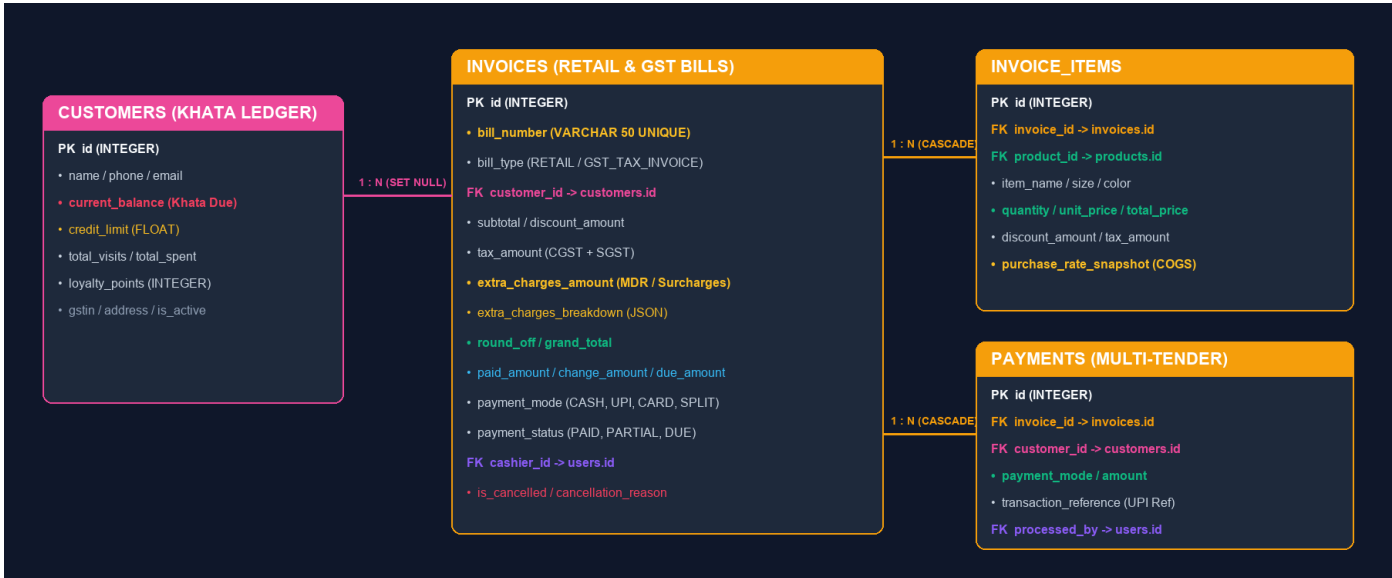


Detailed Schema Specification: Products & Categories

Column Name	Data Type	Attributes	Business Purpose & Description
id	INTEGER	PK Auto-Inc	Unique internal sequence ID for the product.
barcode	VARCHAR(100)	UNIQUE, INDEXED	EAN-13 / Code128 barcode scanned at counter.
sku	VARCHAR(100)	INDEXED	Stock Keeping Unit identifier for inventory audits.
name	VARCHAR(255)	NOT NULL, INDEXED	Product title printed on bills & labels.
category_id	INTEGER	FK -> categories.id	Main department (e.g. Toys, Boys Wear, Girls Wear).
subcategory_id	INTEGER	FK -> subcategories.id	Specific classification (e.g. Frocks, Remote Cars).
purchase_rate	FLOAT	NOT NULL, Default 0.0	Wholesale purchase cost basis used to compute COGS and net profits.
mrp	FLOAT	NOT NULL, Default 0.0	Maximum Retail Price printed on barcode stickers.
selling_price	FLOAT	NOT NULL, Default 0.0	Final retail billing price offered to customers.
current_stock	INTEGER	NOT NULL, Default 0	Live available units on shelf. Decrements on sale, increments on purchase.
damaged_quantity	INTEGER	Default 0	Units damaged during transit or customer trials (excluded from saleable stock).
min_stock_alert	INTEGER	Default 5	Threshold that triggers the AI Reorder Advisor and Low Stock alert.
speed_dial_code	VARCHAR(20)	INDEXED	Quick numeric shortcut for fast keyboard-only billing without scanning.

4. Cluster Deep-Dive: Billing, Invoices & Multi-Tender Payments

This cluster executes high-speed retail checkout transactions. It manages invoice numbering, dynamic MDR & convenience fee calculation (5 configurable rules), split multi-tender settlement (Cash + UPI + Card + Credit Khata), thermal receipt generation, and WhatsApp messaging.



Detailed Schema Specification: Invoices & Invoice Items

Column Name	Data Type	Attributes	Business Purpose & Description
id	INTEGER	PK Auto-Inc	Primary key and internal order sequence.
bill_number	VARCHAR(50)	NOT NULL, UNIQUE	Human-readable invoice number (e.g., INV-1088).
customer_id	INTEGER	FK -> customers.id	Linked customer profile for credit khata & loyalty (NULL for walk-in).
subtotal	FLOAT	NOT NULL, Default 0.0	Gross sum of all line items before discounts, taxes, and surcharges.
discount_amount	FLOAT	Default 0.0	Total bill-level discount given to customer.
tax_amount	FLOAT	Default 0.0	Total GST liability (split 50% CGST + 50% SGST on tax invoices).
extra_charges_amount	FLOAT	Default 0.0	Sum of applied MDR, online convenience, or handling fees.
extra_charges_breakdown	TEXT	JSON Array	Itemized JSON storing exact charge name, rate, and amount per rule.
grand_total	FLOAT	NOT NULL	Net payable bill value = subtotal - discount + tax + extra_charges.
paid_amount	FLOAT	NOT NULL, Default 0.0	Actual currency collected at counter across all tenders.
due_amount	FLOAT	Default 0.0	Unpaid balance automatically debited to Customer Khata ledger.
purchase_rate_snapshot	FLOAT (in items)	NOT NULL	Captures product wholesale rate at the moment of sale for exact P&L.;

5. Cluster Deep-Dive: Vendor Procurement & Supplier Ledger

This cluster manages wholesale inventory procurement, supplier purchase orders, transportation & freight charges, vendor payment ledgers, and the **F10 Fast Buying Notes & Wishlist** engine.



Detailed Schema Specification: Vendors, Purchases & Buying Notes

Column Name	Data Type	Attributes	Business Purpose & Description
purchases.vendor_id	INTEGER	FK -> vendors.id	Supplier supplying the merchandise (Surat, Mumbai, Ahmedabad).
purchases.grand_total	FLOAT	NOT NULL	Total purchase invoice amount including GST & freight charges.
purchases.balance_due	FLOAT	Default 0.0	Pending amount owed to wholesaler (updates vendor current balance).
purchase_items.quantity	INTEGER	NOT NULL	Total pieces ordered. On mark received, auto-increments products.current_stock.
vendor_ledger.balance	FLOAT	NOT NULL	Cumulative debt owed to vendor after every bill or payment.
notes.item_name [F10]	VARCHAR(255)	NOT NULL	Fast wishlist item recorded by store owner for upcoming market trip.
notes.status	VARCHAR(50)	Default 'PENDING'	Workflow status: PENDING -> ORDERED -> RECEIVED.

6. Foreign Key Integrity Matrix & Referential Cascades

Dolly POS enforces strict database-level Foreign Key constraints to maintain absolute relational consistency. The matrix below outlines all 19 foreign key links and their operational cascade rules:

Child Table	Foreign Key Column	Parent Target Table	ON DELETE Action	Integrity Rationale
subcategories	category_id	categories.id	CASCADE	Deleting category removes subcategories.
products	category_id	categories.id	SET NULL	Product records remain safe if category is deleted.
products	subcategory_id	subcategories.id	SET NULL	Product records remain safe if subcategory deleted.
price_history	product_id	products.id	CASCADE	Price logs purged when product is deleted.
invoices	customer_id	customers.id	SET NULL	Sales bills preserved even if customer deleted.
invoice_items	invoice_id	invoices.id	CASCADE	Deleting bill purges line items cleanly.
invoice_items	product_id	products.id	SET NULL	Past sales history preserved with snapshot rate.
payments	invoice_id	invoices.id	CASCADE	Payments cleaned up on bill deletion.
customer_ledger	customer_id	customers.id	CASCADE	Ledger entries deleted with customer profile.
purchases	vendor_id	vendors.id	CASCADE	Purchases tied directly to supplier entity.
purchase_items	purchase_id	purchases.id	CASCADE	Inward line items cleaned up on PO delete.
purchase_items	product_id	products.id	CASCADE	Direct cascade integrity link to product.
vendor_ledger	vendor_id	vendors.id	CASCADE	Payable ledger cleaned up with vendor.
return_orders	original_invoice_id	invoices.id	SET NULL	Return record preserved if original invoice voided.
return_items	return_order_id	return_orders.id	CASCADE	Return items tied to return order header.
expenses	logged_by	users.id	SET NULL	Expense audit preserved if user account removed.
audit_logs	user_id	users.id	SET NULL	Security activity audit trail remains permanent.

7. Database Performance Indexes

To achieve instantaneous response times across hundreds of thousands of transactions, the database incorporates dedicated B-Tree indexes on all high-frequency query paths:

Index Name	Target Table & Columns	Performance Benefit & Query Scope
idx_products_barcode	products (barcode)	Sub-millisecond barcode scanner product resolution at POS counter.
idx_products_speed_dial	products (speed_dial_code)	Instant keyboard numeric shortcut search (e.g. typing '101' for quick billing).
idx_products_name	products (name)	Rapid live product autocomplete dropdown in POS cart.
idx_products_active	products (is_active)	High-speed filtering of live saleable catalog vs archived goods.
idx_products_last_sold	products (last_sold_at)	Dead Stock detection algorithm and AI Reorder advisor fast scans.